Sunlight in a Petrostate: Reimagining Trinidad and Tobago's Energy Future Through Institutional Transformation

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ABSTRACT:
Trinidad and Tobago's reliance on oil and natural gas has entrenched institutional incentives that delay renewable adoption. Despite average Global Horizontal Irradiance of 4.95-5.88 kWh/m²/day, the country maintains one of the lowest renewable energy shares in CARICOM, with 35 GWh deployed. This paper examines why strong solar potential has not translated into distributed generation or a diversified energy strategy. Through political-economy mapping, LCOE analysis, and scenario modeling from 2023 to 2033, it identifies institutional path dependence, hydrocarbon subsidy structures, weak regulatory capacity, and low public energy literacy as dominant constraints. Utility-scale solar is already cost-competitive with gas at $51.44 USD/MWh versus $53.98 USD/MWh in 2023. Accelerated reform generates $99.9 million USD in annual fuel savings by 2033 and reduces CO_2 emissions by 24.2%, while freeing domestic gas for export. Targeted reforms of subsidy realignment, an insulated renewables regulator, scaled solar deployment, and structured public-private partnerships can transform domestic energy supply without sacrificing export competitiveness.

INTRODUCTION:

Trinidad and Tobago occupies a distinctive position in the Caribbean energy landscape. The country is simultaneously the region's largest natural gas producer and exporter and one of its weakest performers in domestic renewable energy deployment (Ministry of Energy and Energy Industries, n.d.). For decades, hydrocarbons have underpinned national fiscal stability, yet this success has entrenched systemic structures that slow renewable adoption relative to neighbouring islands with fewer financial resources but more diversified electricity systems.

While Jamaica, Barbados, and Belize have implemented net-metering, feed-in tariffs, and comprehensive renewable action plans, Trinidad and Tobago maintains minimal installed renewable capacity despite possessing strong solar irradiance and a modern grid infrastructure (Caribbean Centre for Renewable Energy and Energy Efficiency, 2023; Ministry of Energy and Energy Industries, n.d.). The paradox is therefore not one of technical scarcity, but of institutional path dependence of a hydrocarbon-oriented political economy that has historically prioritized export competitiveness and subsidized fossil-fuel consumption over diversification.

This paper argues that **Trinidad and Tobago can scale renewable energy economically and equitably while retaining oil and gas export competitiveness by realigning fiscal and institutional incentives, preserving the hydrocarbon export base while fundamentally transforming domestic energy supply.**

LITERATURE REVIEW:

Historical development of the hydrocarbon economy

Hosein (2021) traces Trinidad and Tobago's hydrocarbon economy to the first commercial oil well at La Brea in 1857, deepening through the rise of LNG and petrochemicals to make the country the world's first LNG exporter outside the Middle East. Karl (1997) explains why this dependence became entrenched by resource rents accruing to the state to create incentives for short-term distributive spending over structural transformation. The eventual collapse of the state-owned oil refinery, Petrotrin, in 2018 was due to deferred maintenance, procyclical fiscal spending, and political undercapitalization; representing the most visible consequence of this dynamic and a cautionary precedent for the electricity sector.

Current Energy System and Resource Potential in Trinidad and Tobago

Natural gas accounts for over 99% of electricity generation (IEA, 2023) through combined-cycle gas plants feeding a centralized grid operated by T&TEC. Blechinger et al. (2015) and Kersey et al. (2021) identify this monopolistic single-buyer structure as a major barrier to renewable expansion, creating slow interconnection processes and no framework for independent power producers. Despite this, Trinidad and Tobago possesses strong solar resources, with GHI averaging 4.95-5.88 kWh/m^2/day and rooftop solar potential estimated at 400-600 MW across commercial and industrial buildings (Regulated Industries Commission, 2019). Boodoo (2025) confirms solar PV's technical viability with system-level LCOEs as low as $0.191 USD/kWh but shows adoption has stalled because capital costs remain high relative to artificially subsidized grid tariffs. Moreover, the 2025 commissioning of the 92 MW Brechin Castle solar project marks a turning point, though Kersey et al. (2021) confirm that scaling requires net-metering frameworks and regulatory reform as prerequisites.

Institutional and political-economic barriers in Trinidad and Tobago

Beyond technical feasibility, Trinidad and Tobago's energy transition is constrained by a dense web of institutional and political-economic barriers. Bee et al. (2025) reinforces this regional pattern, noting that despite successful pilot projects, the Caribbean's overall transition remains slow due to weak regulatory frameworks and fragmented institutional capacity. The core actors, that is, central ministries, T&TEC, Heritage and National Gas Company, multinational corporations, the Oilfields Workers' Trade Union, and domestic financiers each operate under misaligned incentives. Ministries prioritize fiscal stability and low domestic energy prices, T&TEC is incentivized to preserve gas-based supply under its single-buyer model, NOCs and MNCs focus on gas monetization, and unions defend legacy employment. Following Tsebelis (2002), these multiple veto players cluster around subsidy reform, grid access, project approvals, and labour restructuring, systematically impeding policy departure from the status quo, with each actor capable of diluting reform through bureaucratic inertia and political negotiation.

The policy instrument mix further entrenches these barriers. As Dornan and Shah (2016) document across SIDS, fuel and electricity subsidies artificially suppress end-user prices, crowding out renewable investment. This pattern is directly reproduced in Trinidad and Tobago, where subsidized tariffs eliminate the financial incentive for rooftop solar adoption regardless of falling technology costs. Kersey et al. (2021) identify monopolistic utility structures, procurement opacity, and the absence of supportive regulatory frameworks as the most significant institutional constraints to renewable expansion across Caribbean island states, confirming that Trinidad and Tobago's barriers are systemic rather than idiosyncratic. Compounding this, high staff turnover, frequent board changes, and shifting political priorities erode continuity in energy planning. Lockwood et al. (2017) identify such regulatory

discontinuities as a core mechanism of institutional path dependence, through which each disruption resets the reform process and rebuilds the status quo's advantage, ensuring that realigning fiscal and institutional incentives remains a precondition for an equitable renewable energy transition.

Social, behavioral, and equity constraints

Thompson et al. (2024) demonstrate that renewable adoption in Caribbean SIDS remains constrained by limited public understanding and mistrust of institutions. Likewise, Konzen et al. (2024) synthesized 87 studies, confirming that financial barriers like upfront costs and administrative complexity are the most cited obstacles, and that rooftop solar uptake is often regressive without targeted policy design; with higher-income households disproportionately capturing subsidies. Even when subsidies reduce the net cost of installation, lower-income households face prohibitive upfront capital requirements and are less likely to own suitable property or access financing. Additionally, higher-income households are better positioned to navigate administrative processes and benefit from tax credits that require taxable income to claim, compounding the regressive distribution of public support.

Multi-National Corporations (MNC's) & Partnerships

Ragosa and Warren (2019) demonstrate that foreign capital flows where, enabling regulatory conditions exist; without governance reform, renewable resource potential alone will not attract investment. The World Bank's ESMAP (2024) framework identifies blended-finance mobilization and regulatory strengthening as prerequisites for private-sector participation, informing this paper's public-private partnership recommendations.

ANALYSIS:

Economic and environmental costs of the status quo (Levelized Cost Of Energy)

The economic case for transition is compelling. This paper estimates gas LCOE at $53.98 USD/MWh in 2023, rising to $57.06 USD/MWh by 2033 under moderate price growth (π = +1%) and $64.10 USD/MWh under high price growth (π = +3%). Utility-scale solar LCOE, by contrast, sits at $51.44 USD/MWh in 2023, falling to $30.80 USD/MWh by 2033 under a conservative 5% annual cost reduction; a widening advantage of $26.26 USD/MWh. Solar is therefore already cost-competitive with gas, a finding that holds even before accounting for the opportunity cost of domestically priced gas relative to its netback export value. Beyond generation costs, gas-fired generation imposes air quality burdens and water consumption not captured in the LCOE, while the residential electricity subsidy is estimated at $35.6 million USD annually in 2023, rising cumulatively to approximately $398 million USD over 2023–2033 under business-as-usual (BAU).

Policy Roadmap

Drawing on international best practices and existing government frameworks, this roadmap synthesizes a three-phase institutional reform pathway. In the short term (1-3 years), priority actions include subsidy reallocation pilots redirecting a portion of the $2.12 billion TTD fuel subsidy, figured by Imbert (2022) toward rooftop solar financing, streamlined interconnection procedures, and public awareness campaigns. In the medium term (3-7 years), a dedicated renewables regulator established through primary legislation anchors competitive procurement auctions and green bond issuance linked to verified deployment milestones. In the long term (7-15 years), grid modernization proceeds alongside rising renewable penetration, with green hydrogen and ammonia export industries emerging and workforce development programmes

scaling to meet demand. Each phase is stress-tested against adverse conditions including political turnover, higher borrowing costs, and prolonged fiscal constraints.

Model Structure & Inputs

The integrated model spans 2023 to 2033 across six components: baseline demand projection, gas price model, total fuel cost model, LCOE analysis, carbon emissions model, and subsidy/household cost model. Baseline electricity demand is projected using $D_t = D_0(1+g)^t$, where D_0 = 9 467 GWh and g = 0.794%, reaching 10 246 GWh by 2033. The gas price model applies three trajectories to a 2023 baseline of $2.54 USD/MMBtu: declining (π = -0.3%), moderate (π = +1%), and high (π = +3%). Fuel costs combine projected demand with an unweighted average system heat rate of 11 579 MMBtu/GWh derived from the four major generation stations. Three policy scenarios are modeled: BAU (zero renewable deployment, frozen subsidized tariffs), Moderate Reform (15% renewable share by 2033, per the Regulated Industries Commission roadmap), and Accelerated Reform (30% renewable share by 2033, consistent with CARICOM's 2035 targets and the trajectories of Barbados and Jamaica).

Scenario Definitions

Three scenarios define the policy range. The BAU scenario assumes zero renewable deployment, all demand growth met by gas, and tariffs frozen at subsidized levels. The Moderate Reform scenario reaches 15% renewable share by 2033, consistent with the RIC roadmap. The Accelerated Reform scenario reaches 30% renewable share by 2033 through subsidy reallocation, a dedicated renewables regulator, and scaled solar deployment; consistent with CARICOM's 2035 targets and the trajectory of Barbados and Jamaica.

Fuel Cost Results

Under BAU with moderate price growth, total fuel costs rise from $278.4 million USD in 2023 to $332.9 million USD by 2033, and to $405.0 million USD under high price growth. Accelerated reform reduces this to $233.0 million USD under $\pi = +1\%$, generating $99.9 million USD in annual savings; widening to $121.5 million USD under $\pi = +3\%$. Crucially, even under declining gas prices ($\pi = -0.3\%$), accelerated reform still saves $67.5 million USD annually, confirming that the reform dividend is structurally robust rather than contingent on optimistic price assumptions. The policy implication is significant: delaying reform does not reduce risk but increases fiscal exposure, with the reform dividend growing largest under precisely the conditions of greatest fiscal stress.

LCOE Analysis

The LCOE analysis reveals that utility-scale solar is already cost-competitive with gas. Using a capital recovery factor of 0.0937 at an 8% discount rate over 25 years, consistent with industry standard assumptions independent of the 10-year scenario projection window, gas LCOE stands at $53.98 USD/MWh in 2023, rising to $57.06 USD/MWh under $\pi = +1\%$ and $64.10 USD/MWh under $\pi = +3\%$ by 2033. Utility solar LCOE falls from $51.44 USD/MWh in 2023 to $30.80 USD/MWh by 2033 under a conservative 5% annual cost reduction with a gap of $26.26 USD/MWh that widens further under 8% or 10% reduction rates. This finding is particularly significant because it holds even before accounting for the opportunity cost of domestically priced gas relative to its netback export value; under a reformed pricing regime, solar's competitive advantage becomes even more pronounced.

Rooftop solar presents a more nuanced picture. Falling from $91.64 USD/MWh in 2023 to $54.87 USD/MWh by 2033, it remains more expensive than utility solar throughout the

projection window. The relevant comparison for households, however, is not generation cost but avoided retail tariff as the model shows rooftop solar reaching near-breakeven only by 2033. This carries a critical sequencing implication as households cannot be expected to drive distributed solar adoption under current subsidized tariffs regardless of falling technology costs. The financial incentive for household investment is a policy choice, not a market outcome, contingent entirely on whether tariff reform precedes or accompanies solar deployment programmes.

Fiscal Balance, Household Costs, and Emissions

Under accelerated reform, average household electricity bills rise from $17.90 USD/month in 2023 to $24.74 USD/month by 2033. This increase of approximately $0.69 USD/month per year is both fiscally manageable and politically defensible; provided means-tested relief protects lower-income households. On emissions, BAU produces a steady rise from 5.30 $MtCO_2$ in 2023 to 5.74 $MtCO_2$ by 2033 resulting in an 8.3% increase driven entirely by demand growth. Moderate reform reverses this to 4.88 $MtCO_2$ (-7.9%), while accelerated reform achieves 4.02 $MtCO_2$ (-24.2%), avoiding 1.72 $MtCO_2$ annually relative to BAU. Sensitivity analysis across demand growth trajectories (0.3%, 0.794%, and 1.5%) confirms that emissions rise under all three BAU trajectories and even low growth produces an increase from 5.30 to 5.46 $MtCO_2$, flatly contradicting any argument that demand-side management alone can stabilize emissions. Structural reform of the generation mix is not merely preferable but necessary. Conversely, under accelerated reform, higher demand growth makes the emissions and fuel cost savings larger in absolute terms, strengthening rather than weakening the economic case for transition.

Sensitivity Analysis

Sensitivity analysis confirms robustness across all tested assumptions across three dimensions: gas price trajectories, solar cost reduction rates, and demand growth rates. Figure 1 illustrates fuel cost savings from accelerated reform range from $67.5 million USD under the declining price scenario (π = -0.3%) to $121.5 million USD under the high price scenario (π = +3%) by 2033, confirming that the reform dividend grows largest under precisely the conditions of greatest fiscal stress. This asymmetry means delay increases rather than reduces the economic argument for transition.

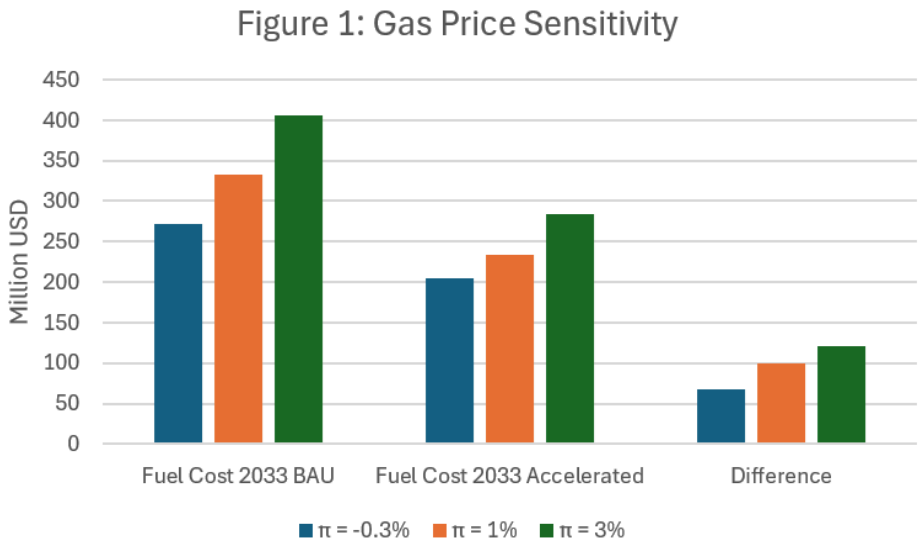
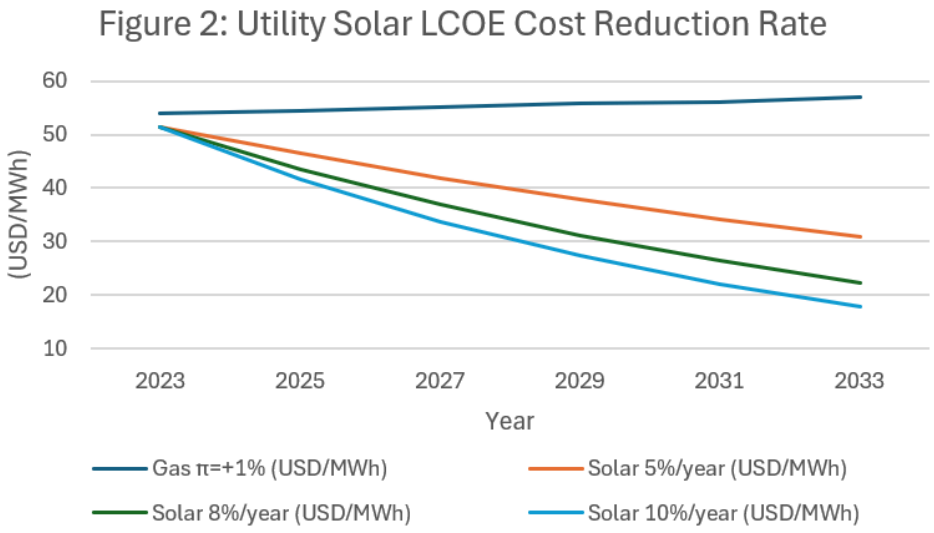
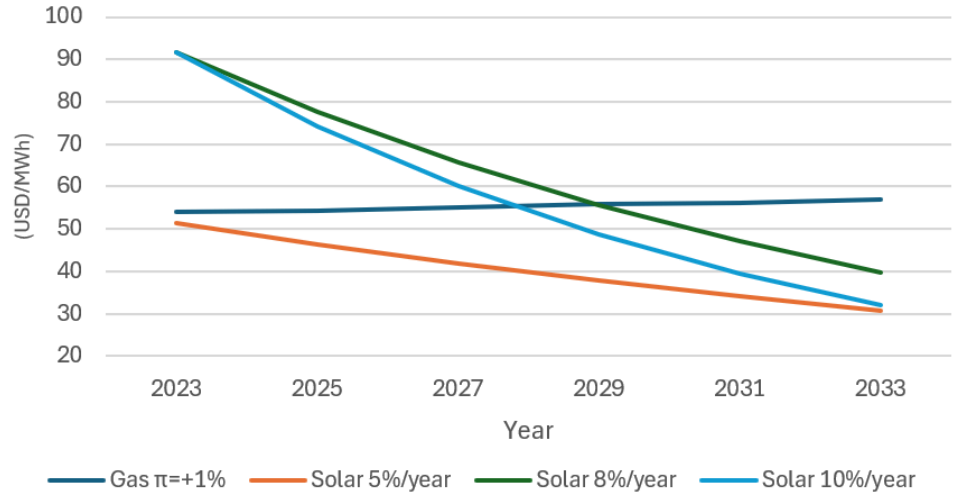


Figure 2 visualizes the utility-scale solar LCOE sensitivity against three cost reduction rates, conservative (5%/year), moderate (8%/year), and aggressive (10%/year), compared to the gas trajectory under π = +1%. Critically, utility solar is cheaper than gas at the 2023 baseline before the projection period begins. Under the conservative rate, solar falls to $30.80 USD/MWh by 2033 versus gas at $57.06; a $26.26 USD/MWh gap. Under moderate and aggressive rates, the gap widens to $34.72 and $39.12 USD/MWh respectively. Every year of delayed utility-scale deployment therefore represents a growing opportunity cost.

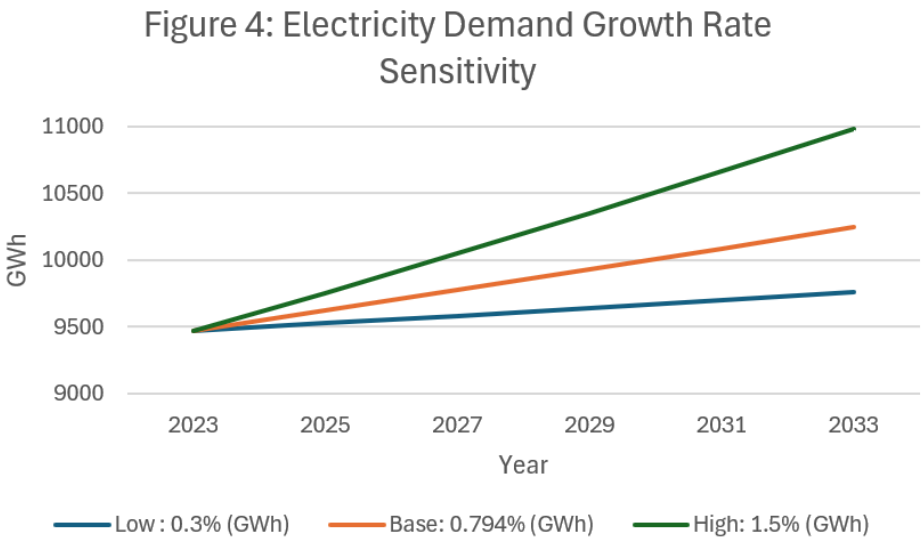


Contrastingly, Figure 3 illustrates the rooftop solar LCOE cost reduction rate sensitivity against the gas LCOE trajectory under $\pi = +1\%$. Three solar cost reduction rates are tested: conservative (5%/year), moderate (8%/year), and aggressive (10%/year). Under all three rates, rooftop solar LCOE declines steadily from its 2023 baseline of $91.64 USD/MWh to $54.87, $39.81, and $31.95 USD/MWh respectively by 2033. Critically, gas LCOE under $\pi = +1\%$ rises gradually from $53.98 to $57.06 USD/MWh over the same period. The crossover point where rooftop solar becomes cheaper than gas occurs approximately at 2031 under the 8% rate and 2029 under the 10% rate, while under the conservative 5% rate rooftop solar approaches but does not cross below gas within the projection window. This finding reinforces the sequencing argument that rooftop solar competitiveness is not guaranteed within the 2023-2033 horizon under conservative assumptions, making tariff reform and upfront financing support essential policy complements rather than optional additions.

Figure 3: Rooftop Solar LCOE Cost Reduction Rate

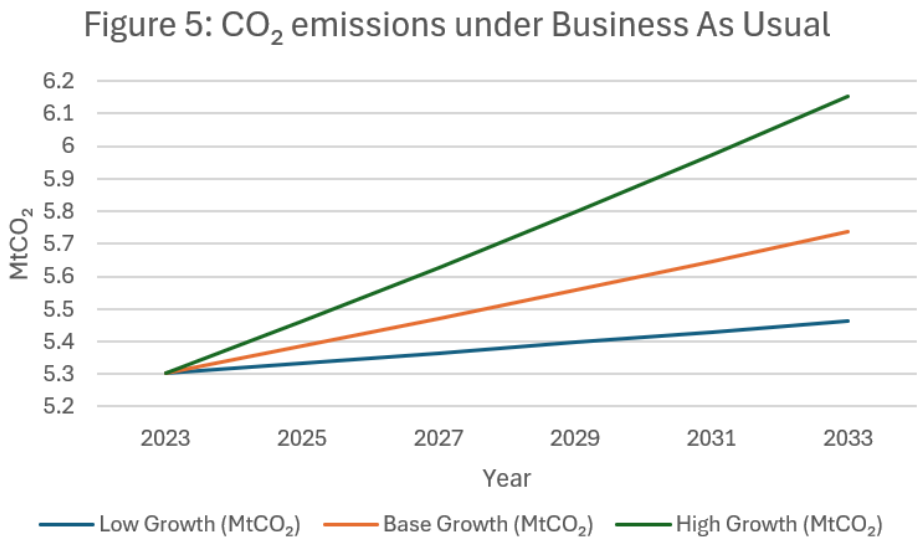


Continuing, Figure 4 presents three electricity demand growth trajectories, low (0.3%), base (0.794%), and high (1.5%) from 2023 to 2033. The divergence between scenarios widens progressively over time, with total demand ranging from 9 755 GWh under low growth to 10 987 GWh under high growth by 2033, compared to the base case of 10 246 GWh. This spread has direct implications for the emissions and fuel cost models as under high demand growth, BAU emissions reach 6.15 $MtCO_2$ by 2033 versus 5.46 $MtCO_2$ under low growth. Importantly, emissions rise under all three trajectories in the BAU scenario confirming that demand-side management alone, even if growth slows significantly, cannot stabilize emissions without structural reform of the generation mix. Conversely, under accelerated reform the higher demand growth scenario makes the emissions and fuel cost savings even larger in absolute terms, strengthening rather than weakening the economic case for transition.



Lastly, Figure 5 presents CO_2 emissions trajectories under BAU across all three demand growth scenarios from 2023-2033. Two findings are particularly significant. First, the divergence between scenarios accelerates over time as the spread between low and high growth trajectories widens from approximately 0.26 $MtCO_2$ by 2027 to 0.69 $MtCO_2$ by 2033, meaning delayed reform under high demand growth produces disproportionately larger emissions outcomes than a linear extrapolation would suggest. Second, and most importantly, all three trajectories rise without exception. Even under low growth, BAU emissions increase from 5.30 to 5.46 $MtCO_2$ by 2033, flatly contradicting any argument that demand-side management alone could stabilize emissions. As long as the generation mix remains entirely gas-dependent, any positive demand growth produces rising emissions, making structural reform of the generation mix not merely preferable but necessary. Against the high growth upper bound of 6.15 $MtCO_2$, accelerated reform's projected outcome of 4.02 $MtCO_2$ represents an avoided gap of 2.13 $MtCO_2$ annually by 2033; a figure that grows further if demand accelerates. Together with Figure 2's utility solar cost analysis, Figure 5 completes the quantitative argument that the reform case is self-reinforcing across the full sensitivity space. Economic benefits are largest when gas prices are highest,

solar's cost advantage is largest when learning rates are fastest, and environmental benefits are largest when demand growth is strongest.



Together, these results demonstrate that the economic and environmental case for transition is not contingent on any single favorable assumption but is driven by the convergence of falling solar costs, rising gas price risk, growing fiscal subsidy burden, and compounding emissions under hydrocarbon dependence. The robustness of this finding across the full assumption space is the strongest quantitative support this paper offers for its central thesis.

The modeling results demonstrate a compelling economic and environmental case for renewable energy transition, yet translating analytical findings into implemented policy requires confronting Trinidad and Tobago's political economy. The sensitivity analysis is not merely a technical robustness check, it directly informs the political economy risks, mitigation strategies, and accountability mechanisms that determine whether reform succeeds.

Political Economy Risks & Mitigation

Vested-interest resistance is the most immediate constraint. The hydrocarbon coalition, that is, Heritage, NGC, the OWTU, and industrial consumers benefiting from subsidized gas occupies key veto points, enabling technically sound reform proposals to be diluted through bureaucratic inertia. Under declining gas prices, when the annual reform dividend falls to $67.5 million USD, vested interests can argue urgency is overstated, making the political economy case for reform most vulnerable precisely when incumbents feel least threatened. Electoral cycles compound this as Karl's (1997) analysis of petrostates documents the consistent tendency to expand subsidies and defer structural reform ahead of elections; a dynamic Hosein (2021) traces directly in the undercapitalization and deferred maintenance that produced the Petrotrin collapse.

Three mitigation strategies address these risks. First, credible commitment devices through establishing the renewables regulator through primary legislation with fixed-term appointments and judicial review insulates reform from electoral cycles. The sensitivity analysis shows the reform dividend is robust across all gas price trajectories ($67.5 - 121.5 million USD), providing a credible justification for regulatory continuity regardless of the price environment. Second, phased compensation mechanisms like gradual tariff adjustment of $0.69 USD/month per year is politically defensible; means-tested electricity credits protect lower-income households and structured transition packages convert union opposition into stakeholder support. Third, stakeholder coalition building can communicate the fuel cost savings to commercial and industrial sectors which bear high energy costs and gain from lower prices; creating a countervailing coalition to hydrocarbon vested interests.

Monitoring and Accountability

Reform progress should be tracked through annual energy transition scorecards similar to the Trinidad and Tobago Energy Report Card (2021). Monitoring renewable share against the 15% and 30% targets, fuel cost savings against the $99.9 million USD benchmark, and CO_2 emissions against the 4.02 $MtCO_2$ pathway, with deviations triggering automatic institutional review. Performance covenants embedded in green bond issuance link coupon payments to verified capacity additions, creating a financial accountability mechanism independent of political cycles, while a quarterly multi-stakeholder energy transition council provides adaptive governance against the medium- and long-term stress risks identified in the reform roadmap.

CONCLUSION:

This paper has argued that Trinidad and Tobago can scale renewable energy economically and equitably while retaining oil and gas export competitiveness by realigning the fiscal and institutional incentives that currently lock the country into hydrocarbon dependence. Subsidy structures that suppress the financial case for solar, a single-buyer grid model limiting distributed generation, weak regulatory capacity, and a political economy favouring short-term rents over long-term energy security are the binding constraints, but they are not insurmountable.

The modeling demonstrates that utility-scale solar is already cost-competitive with gas, accelerated reform generates $99.9 million USD in annual fuel savings by 2033, and CO_2 emissions fall 24.2%, all while freeing domestic gas for higher-value export. Regarding policy, the three highest-impact policy actions are, immediate utility-scale solar procurement (the economic case exists now, independent of broader institutional reform), tariff reform before rooftop solar incentive programmes (household adoption incentives fail under current subsidized

tariffs), and establishment of a legally insulated renewables regulator in the short term to anchor reform against electoral cycles.

Limitations include data constraints on station-level generation shares and the exclusion of grid modernization costs. Future research should incorporate dynamic grid modeling, empirical adoption surveys, and a full social cost-benefit analysis including health and ecosystem externalities.

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